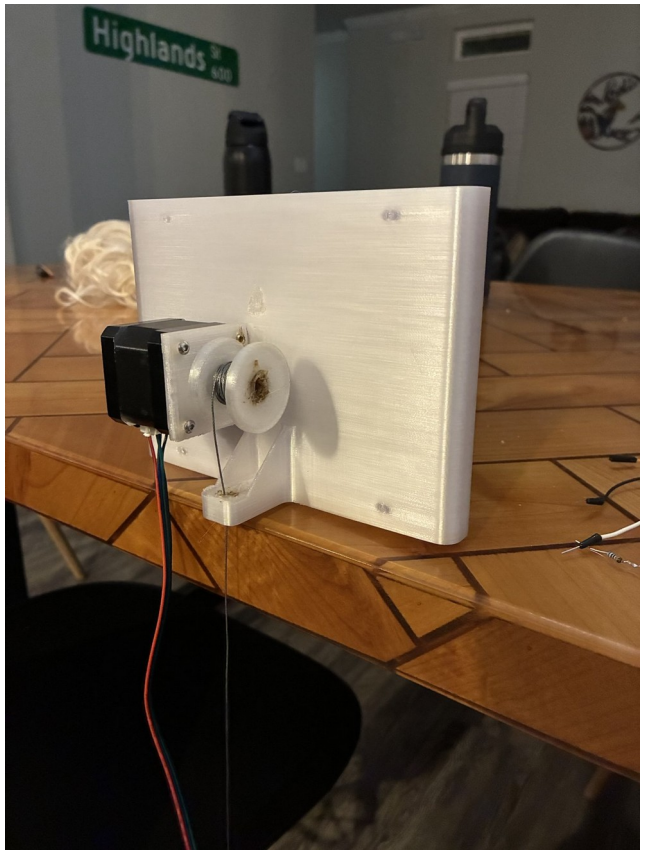
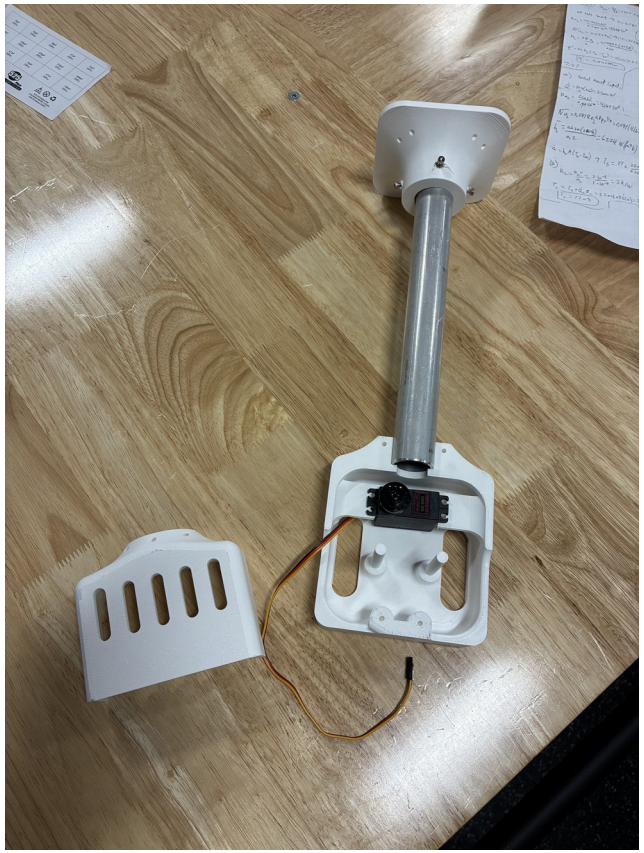
# RoomCleaner

A cable-driven robot that picks laundry up off the floor

Jasper Buntinx
Mechanical Engineering, Texas A&M University
October 27, 2026



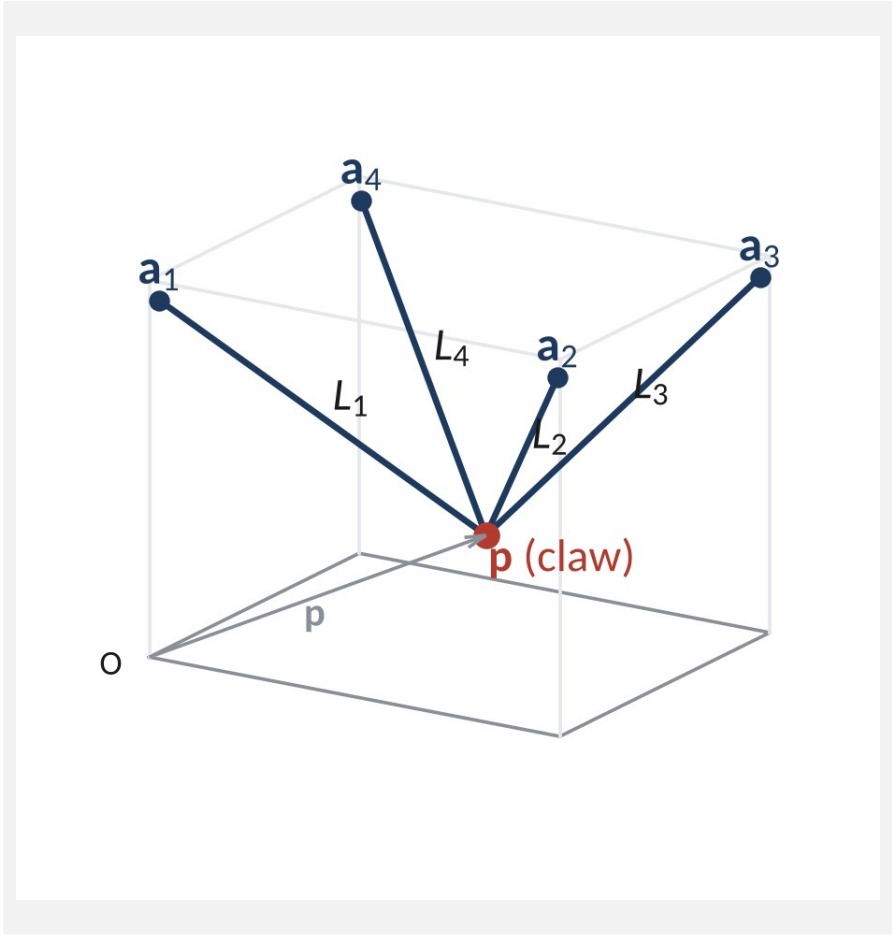
# About Me

- B.S. Mechanical Engineering, Texas A&M University, *[graduation month and year]*
- Built EcoFil, a filament recycler with a shredder, dryer and extruder
- Metal sculptures for the Austin Creek Show, made with fiber laser cutting and welding
- Outside of class: *[hobbies and interests]*
- Why this project: *[one sentence]*

*[Photo of you building: EcoFil, a sculpture, or the workshop]*

# Agenda

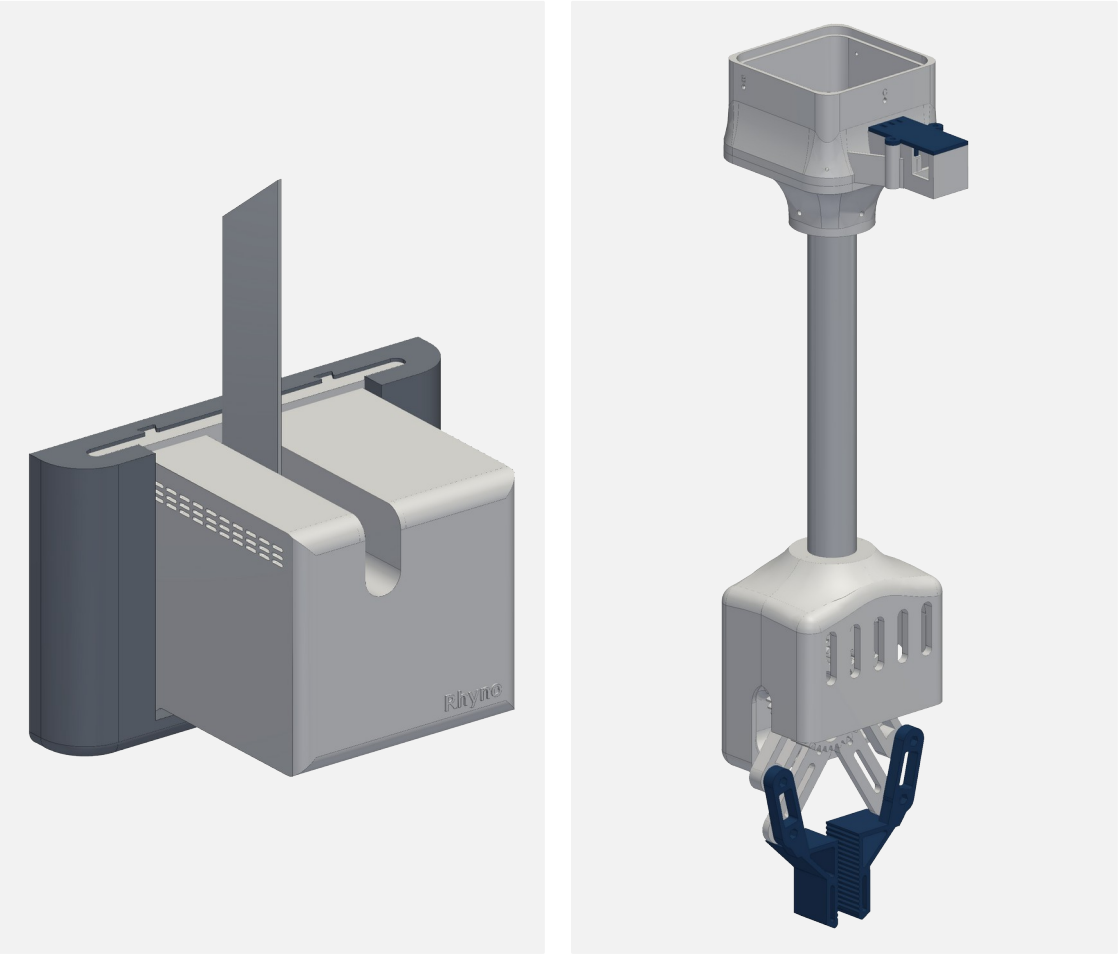
1. Project goal, scope and timeline
2. Design: targets, trade-offs and key calculations
3. Build: printed parts, machining and electronics
4. Test: first system test and thermal verification
5. Good and poor decisions, and lessons learned




Cables from the ceiling anchors $a_1$ to $a_4$ to the claw at p

# Project Goal

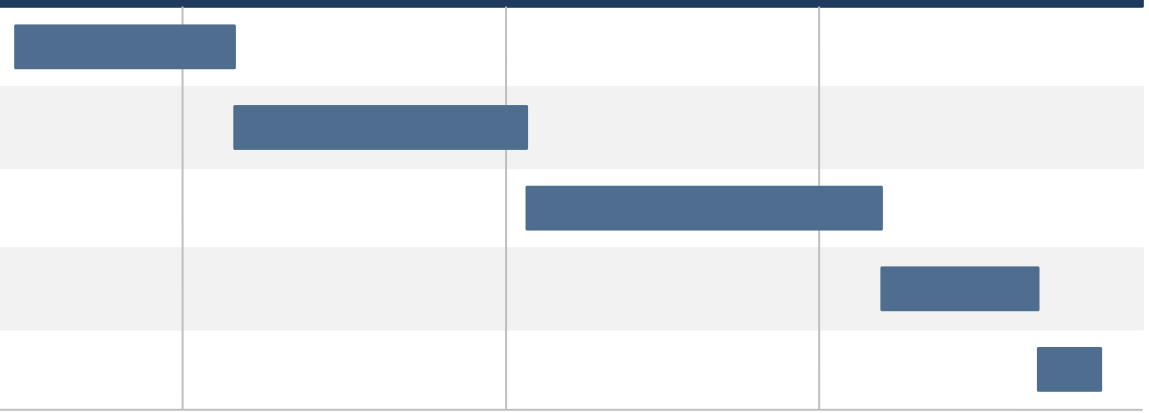
- Pick up laundry anywhere in an 11 × 9 ft room and drop it in a hamper
- Four ceiling-mounted winches suspend one claw on cables
- Design load: jeans, 0.9 kg, at 2 m/s² acceleration; first test load is 1 lb
- 3D-printed PETG (polyethylene terephthalate glycol) parts on a hobby budget
- Success: one garment delivered to the hamper with no human input



Final design (CAD): wall station, left; claw, right

# Scope and Timeline

| Phase | Dates | July | August | September | October |
|---|---|---|---|---|---|
| Concept and requirements | Jul 16 – Aug 5 | | | | |
| Generation 1: build and bench test | Aug 6 – Sep 2 | | | | |
| Redesign: sensors and wiring | Sep 3 – Oct 6 | | | | |
| Generation 2 and final design | Oct 7 – Oct 21 | | | | |
| Build and first system test | Oct 22 – Oct 27 | | | | |



## My work (solo project)

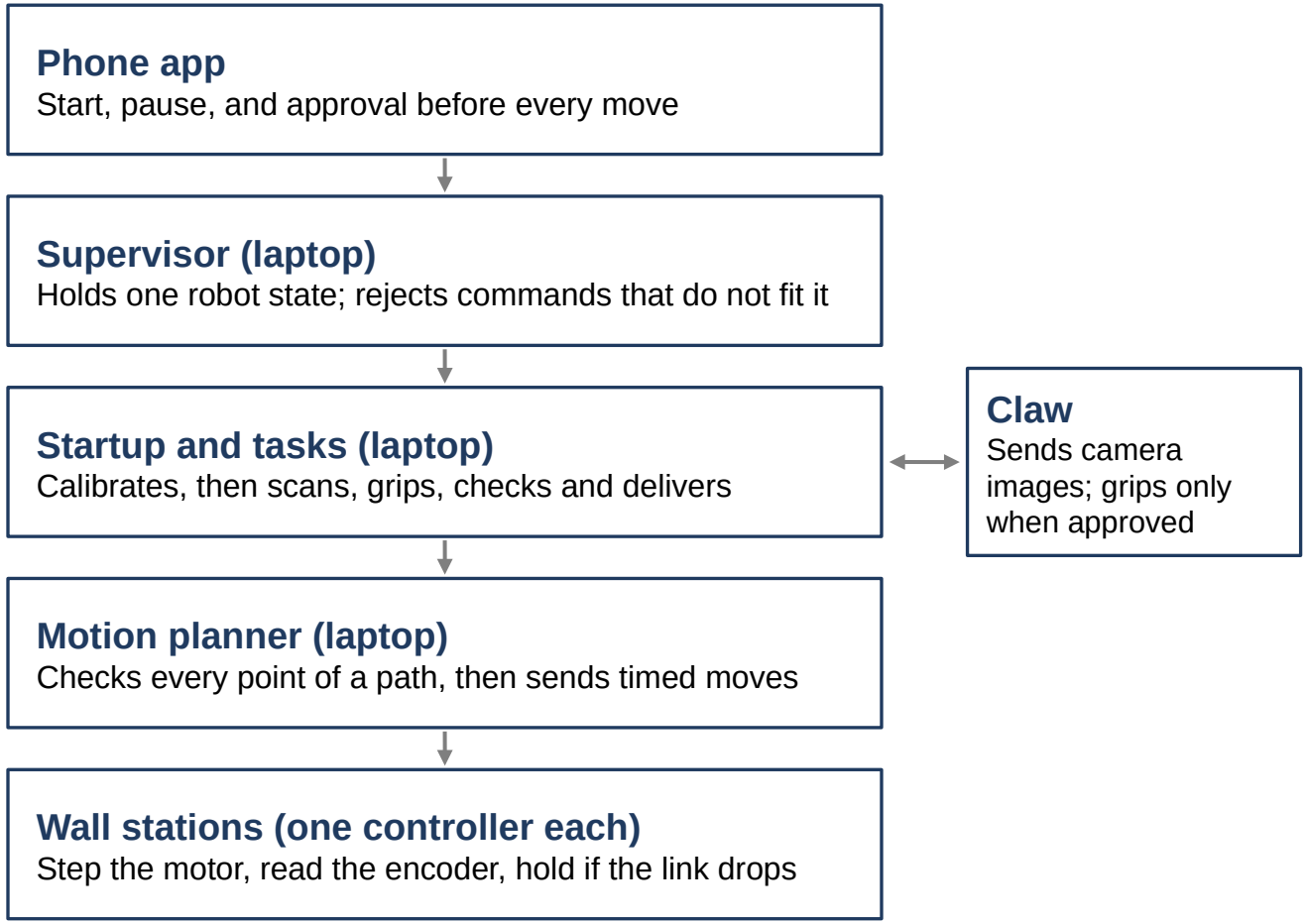
- Requirements, calculations and test plans
- All CAD (computer-aided design) in SolidWorks
- 3D printing, fit testing and machining the heat fin
- Wiring, soldering and electronics bring-up
- Control software: *[confirm the parts you wrote]*

## Purchased or external

- Stepper motors, drivers and controller boards
- Servo, camera, battery and encoder boards
- Garment detection: YOLO-World, an open-source model
- Shop access or mentors: *[if any]*

# How It Works

- Four wall stations, each with its own winch, encoder and controller
- One claw with a gripper, camera and battery, hung on four cables
- Laptop plans each move and checks every cable tension before it runs
- Phone app starts, pauses and approves each move
- Software simulated end to end; hardware test this week




Commands flow down; status and camera images flow up.

# Design Targets

| Parameter | Target | Current value | Status |
|---|---|---|---|
| **Lift capacity** | 1.76 kg: claw, battery and jeans | 1.56 kg (calculated) | **Not met** |
| **Claw mass** | 0.45 kg | 0.86 kg with battery (estimated); *[weighed]* | **Not met** |
| **Motor temperature** | Below 60 °C (PETG limit) | 58 °C prototype fin; 45–49 °C with 1/8 in fin | Marginal |
| **Working area** | Full 11 × 9 ft floor | All but 0.2 m along the walls | Met in design |
| **Wiring** | One cable per station | One 12 V pair; 0.51 V drop on the longest run | Met |
| **Positioning accuracy** | 2 mm per cable | 0.7 mm (calculated) | To be tested |
| **Speed** | 50 mm/s at 2 m/s² | Not yet run | To be tested |
| **Cost** | Hobby budget | *[total cost]* | *[fill in]* |

Safety is a fixed constraint and is not traded against any of these targets.

# Design Trade-offs

| Design change | Benefit | Cost | Chosen setting |
|---|---|---|---|
| **Higher motor current** | 41 % more pull at 1.5 A | Heat rises with current squared: 2.0 W to 4.05 W | 1.5 A, vented shell and heat fin |
| **Stronger claw** | Gear safety factor from 0.4 to 2.0 | More claw mass, less payload | Module 1.5 gears |
| **Larger room** | More floor covered | Longer wires, more voltage drop | Only 12 V power runs to the stations |
| **Working closer to the ceiling** | More working volume | Cable tension rises from 12 N to 32 N | Stay 0.4 m below the anchors |
| **Smaller spool radius** | More pull per amp | Slower line speed | 15 mm core: 34 N empty, 25 N full |
| **Larger safety margins** | No skipped steps or wall contact | Less usable force and floor area | Force ÷ 1.5; 0.2 m from walls |

Key coupling: a heavier claw needs more pull, more pull needs more current, and heat rises with the square of the current.

# Cable Tension

- Cables can only pull, so tension must stay positive
- Tension rises sharply near the ceiling
- 32 N per cable at 0.15 m below the anchors; 8.3 N at 0.6 m
- Operating limit: 0.4 m below the anchors

**Cable tension at the operating limit**

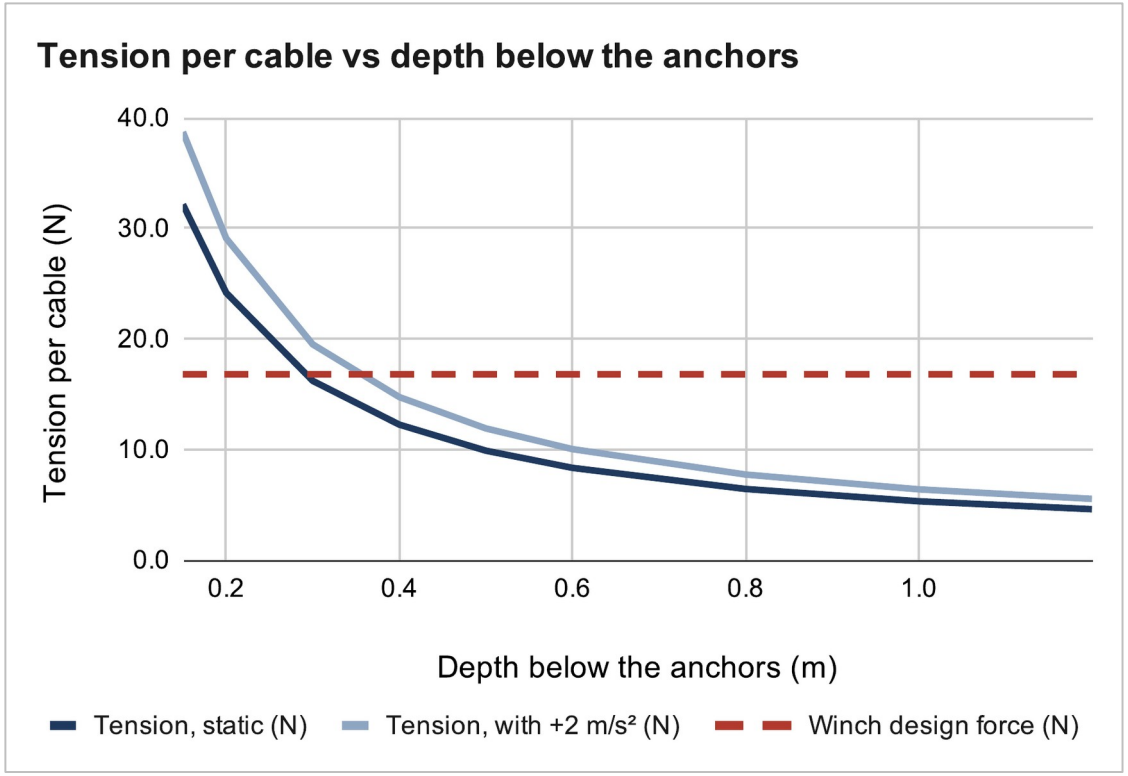
$$T = W / (4 \sin \varphi), \quad \sin \varphi = d / \sqrt{(d^2 + D_h{}^2)}$$

$$\sin \varphi = 0.4 / \sqrt{(0.4^2 + 2.17^2)} = 0.181$$

$$T = 8.87\ \text{N} / (4 \times 0.181) = \mathbf{12.2\ N}$$

$$\text{With } 2\ \text{m/s}^2\text{: } 12.2 \times 1.20 = \mathbf{14.7\ N} < 16.8\ \text{N}$$

**Where:** W = (0.45 + 0.454) kg × 9.81 m/s² = 8.87 N; d = 0.4 m below the anchors; $D_h$ = 2.17 m, room center to corner; 16.8 N = winch design force




Source: RoomCleaner engineering calculations, “Cable tension” (calculated values)

# Winch Pull

- NEMA 17 (National Electrical Manufacturers Association) stepper with a printed spool
- Driver current raised to the rated 1.5 A (was 1.06 A)
- Pull: 34 N with an empty spool, 25 N when full
- Luggage-scale pull test to confirm the torque

**Winch design force at 1.5 A**

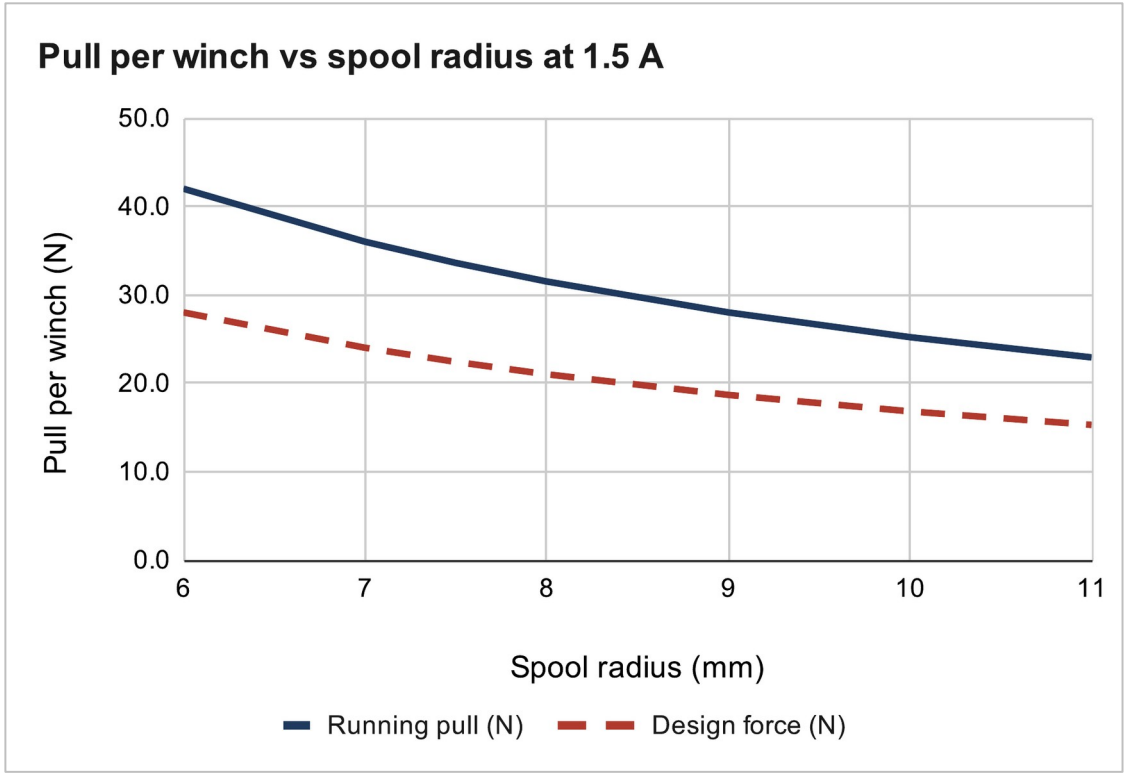
$$I = V_{ref} / (8\,R_s) = 1.2\ V / (8 \times 0.1\ \Omega) = \mathbf{1.5\ A}$$

$$\tau_{run} = 0.7 \times 0.36\ N{\cdot}m = 0.252\ N{\cdot}m$$

$$T = \tau / r = 0.252\ N{\cdot}m / 0.010\ m = \mathbf{25.2\ N}$$

$$F_{design} = 25.2\ N / 1.5 = \mathbf{16.8\ N}$$

**Where:** $V_{ref}$ = driver reference voltage; $R_s$ = sense resistor; 0.36 N·m = holding torque at 1.5 A; 0.7 = running/holding ratio (assumed); r = 10 mm full spool; 1.5 = skip margin




Source: RoomCleaner engineering calculations, “Winch pull” (calculated values)

# One Controller per Winch

- Generation 1 drove all four motors from one central board
- Adding sensors at each winch required 10 wires per corner
- Moved to a separate controller in each station
- Total wire reduced from 820 ft to 164 ft

**Wire length and timing error**

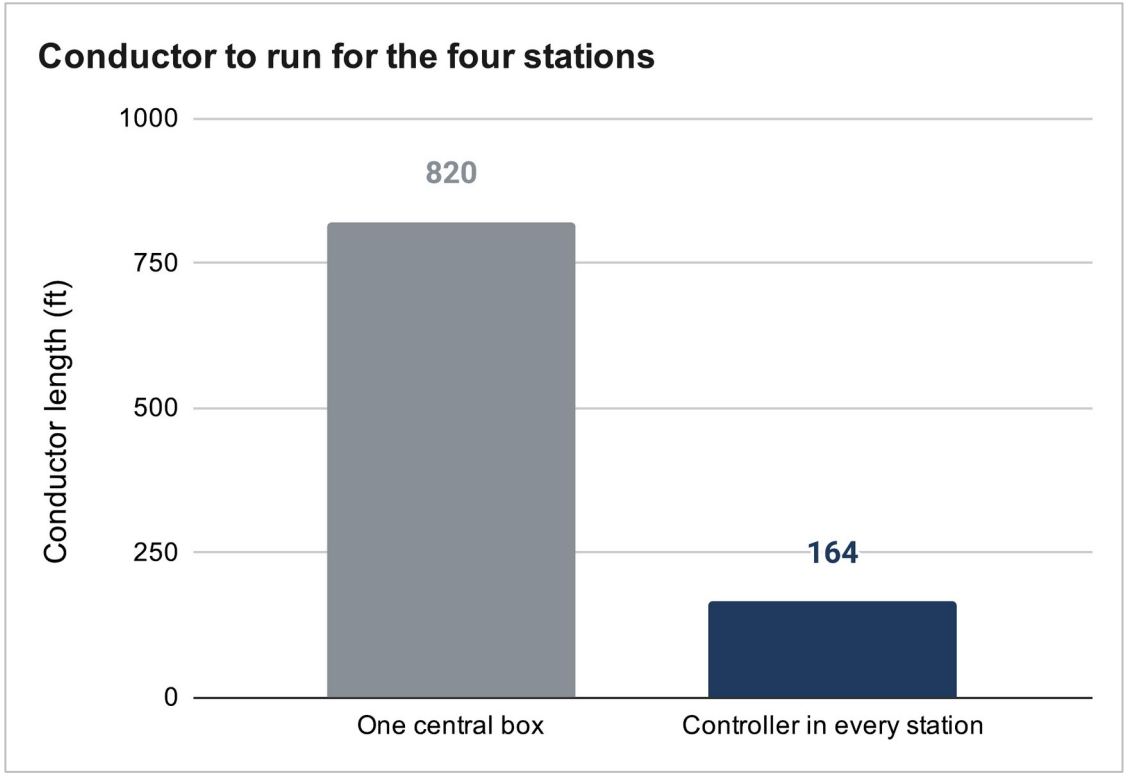
Wire = $n \times \Sigma L$

Central board: $10 \times 82$ ft = **820 ft**

Controller per station: $2 \times 82$ ft = **164 ft**

Timing error: $\Delta x = v\,\Delta t = 50\ \text{mm/s} \times 50\ \text{ms}$ = **2.5 mm**

**Where:** n = conductors per corner (motor 4, switch 2, encoder 4); ΣL = 6 + 21 + 23 + 32 ft of cable routes; v = line speed; Δt = clock offset between stations




Source: RoomCleaner engineering calculations, “Wiring” (calculated values)

# Motor Cooling

- Sealed PETG shell traps motor heat
- PETG creeps above 60 °C under load
- Aluminum fin added through the top of the shell
- 90-minute hold test at 1.5 A planned

**Motor heat and temperature at 1.5 A**

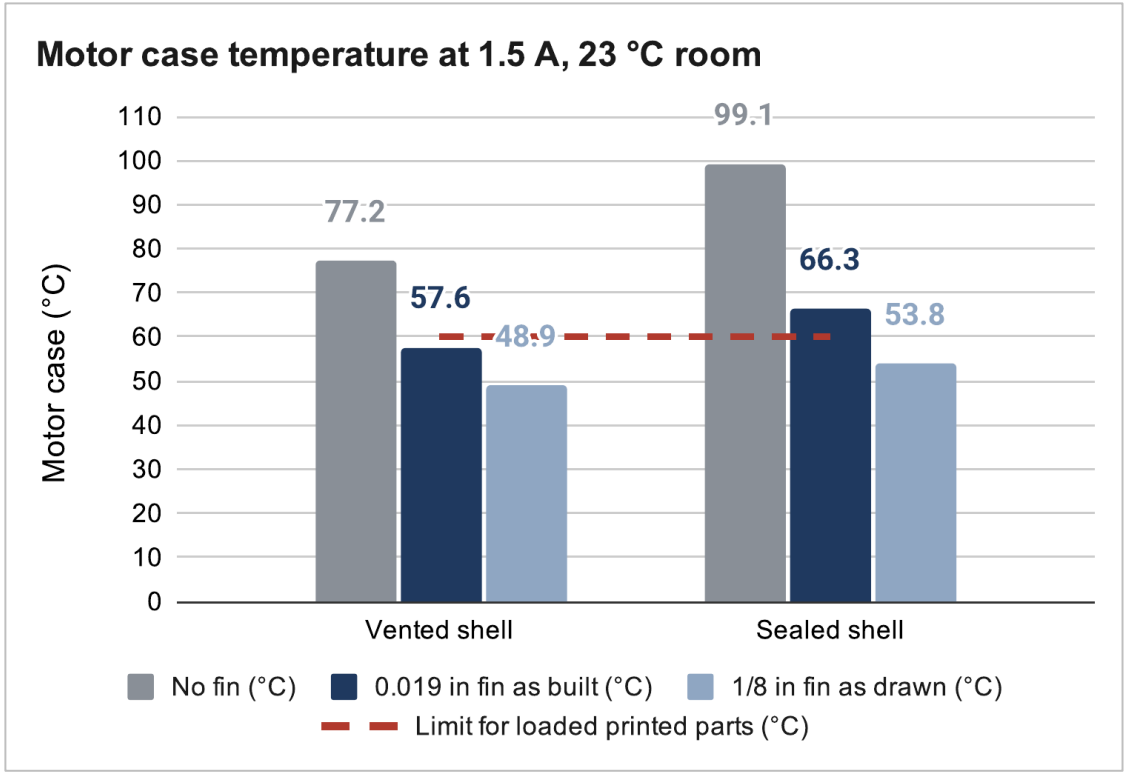
$$P = I^2 R_{coil} = (1.5\ A)^2 \times 1.8\ \Omega = \mathbf{4.05\ W}$$

$$1/R_{th} = 1/R_{shell} + 1/R_{fin},\quad \Delta T = P \times R_{th}$$

$$0.019\text{ in fin: } R_{fin} = 23.1\ K/W \rightarrow \mathbf{57.6\ °C}$$

$$1/8\text{ in fin: } R_{fin} = 11.9\ K/W \rightarrow \mathbf{48.9\ °C}$$

**Where:** $R_{coil}$ = 1.8 Ω per phase (assumed); $R_{shell}$ = 12 + 1 K/W, motor to shell air to room; room at 23 °C; 1.5 W of electronics in the shell; limit 60 °C




Source: RoomCleaner engineering calculations, “Motor temperature” (calculated values)

# Heat Fin Sizing

- 0.019 in prototype fin reaches 95 % of its capacity by 136 mm
- 1/8 in design fin is 74 % efficient at 200 mm
- Built length of 200 mm, limited by ceiling clearance

**Fin length and efficiency**

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

0.019 in: $m = \sqrt{(2 \times 7 / (160 \times 0.00048))} = 13.5\ /m$

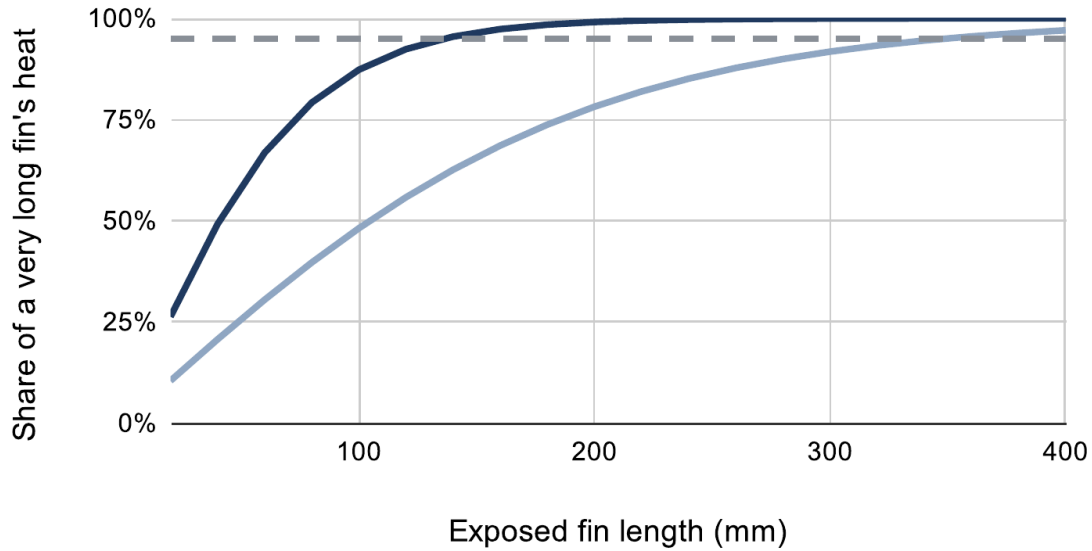
$L_{95} = 1.83 / 13.5 =$ **136 mm**

1/8 in: $m = 5.25\ /m$, $L_{95} =$ **349 mm**

$\eta = \tanh(mL) / (mL) = \tanh(1.05) / 1.05 =$ **0.74**

$R_{fin} = 1 / (7 \times 0.74 \times 0.0176) =$ **10.9 K/W**

**Where:** h = 7 W/m²·K, still air; k = 160 W/m·K, aluminum; t = fin thickness; L = 200 mm; A = 2 × 44 mm × 200 mm = 0.0176 m²; $R_{fin} = 1 / (h\,\eta\,A)$




Source: RoomCleaner engineering calculations, “Heat fin” (calculated values)

# Homing Method

- Original plan: a bead on each cable trips a switch at the outlet
- Cables could not get shorter than their homing length
- Replaced with camera calibration on a printed floor board
- Full room now reachable; beads kept as end stops

**Reachable area with bead homing**

$$L_i(\mathbf{p}) \geq L_{i,\text{home}} = \|\mathbf{a}_i - \mathbf{p}_{\text{home}}\|$$

Home 1 m above the floor: **5 %** reachable

Home 2 m above the floor: **29 %** reachable

**Where:** $a_i$ = anchor i; p = claw position; reachable = every cable at or above its home length, inside the 0.2 m wall and 0.4 m ceiling margins




Source: RoomCleaner engineering calculations, “Bead homing” (calculated values)

# Claw Design

- Generation 1 used five TPU (thermoplastic polyurethane) fingers
- Fingers could not pinch thin fabric and were 4.6 times over the mass budget
- Final design: geared four-bar with parallel jaws and a 53 mm opening

**Grip force and gear stress at servo stall**

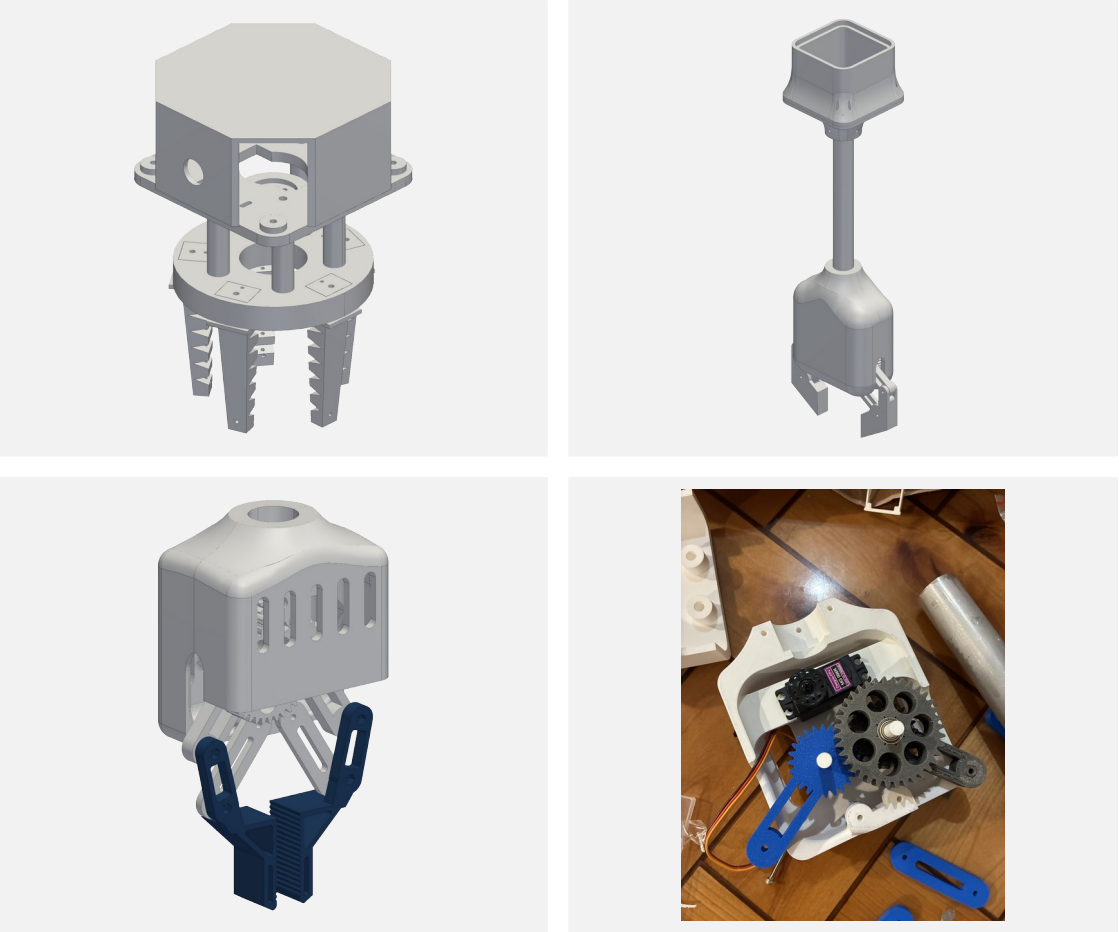
$$N \geq W / (2\mu) = 4.45\ \text{N} / 0.6 = \mathbf{7.4\ N}$$

$$F_t = 2\tau / d = 2 \times 1.078 / 0.027 = 79.9\ \text{N}$$

$$\sigma = F_t / (b\ m\ Y) = 79.9 / (15.7 \times 1.5 \times 0.309)$$

$$\sigma = 11.0\ \text{MPa},\ SF = 22.5 / 11.0 = \mathbf{2.05}$$

**Where:** N = grip force per jaw; W = 1 lb test load; μ = 0.3; τ = 1.078 N·m servo stall torque; d = 27 mm pitch diameter; b = 15.7 mm face width; m = 1.5 mm module; Y = 0.309 Lewis factor; SF = safety factor



Left to right, top to bottom: generation 1 fingers, v6 clamp, final four-bar (CAD), printed gear train

# Weight Budget: Open Issue

- Lift capacity at 1.5 A is 1.56 kg
- Claw with battery estimated at 0.86 kg *[weighed mass]*
- 1 lb test load fits; jeans exceed capacity by 0.2 kg
- Options: a lighter claw or a 3:1 gear reduction on the winch

**Lift capacity against the required mass**

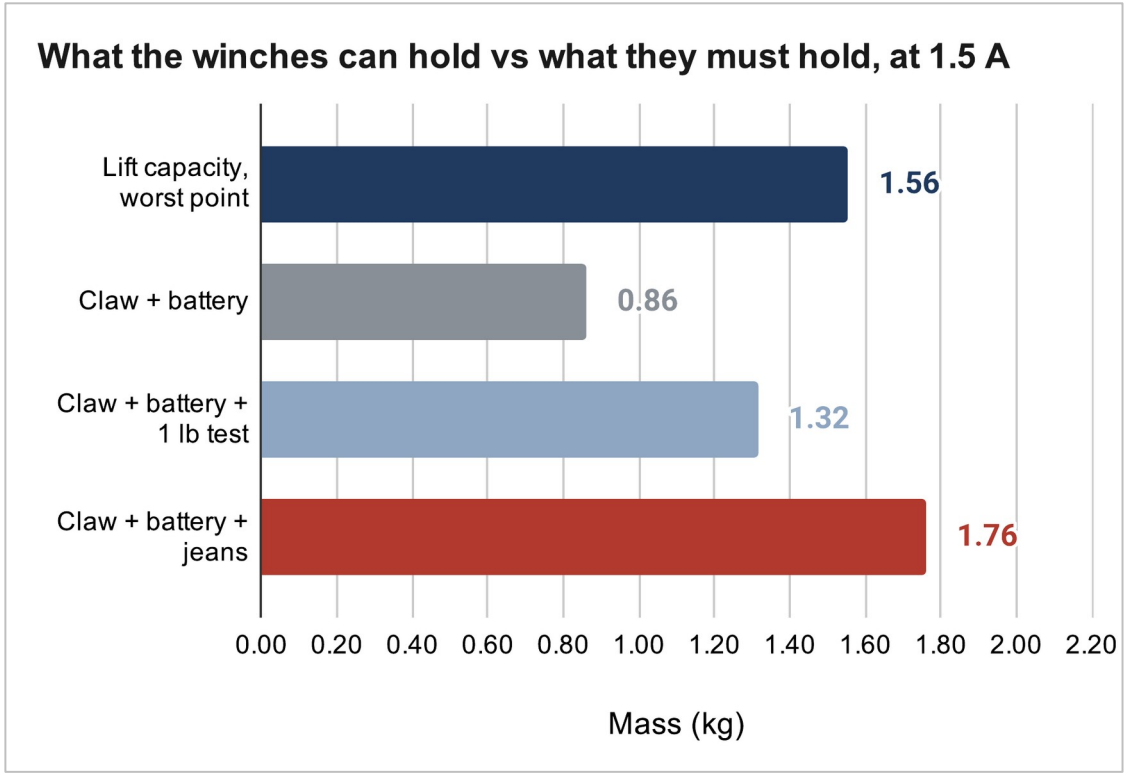
$$m_{max} = F_{design} / (k\,g)$$

$$= 16.8\ N / (1.101 \times 9.81\ m/s^2) = \mathbf{1.56\ kg}$$

$$m_{claw} = 0.55 \times 0.927 + 0.198 + 0.055 = \mathbf{0.76\ kg}$$

$$\text{Required: } 0.76 + 0.10 + 0.90 = \mathbf{1.76\ kg} > 1.56\ kg$$

**Where:** k = worst cable tension per unit weight over the room, with 2 m/s²; 0.55 = print fill; 0.927 kg = solid PETG from CAD; 0.198 kg tube; 0.055 kg servo; 0.10 kg battery; 0.90 kg jeans




Source: RoomCleaner engineering calculations, “Lift budget” (calculated values)

# Printed Part Fits

- PETG printed on a Creality K1 at 0.2 mm layers
- Printed holes come out undersize by at least 0.2 mm
- The spool needed three prints to fit the motor shaft
- Fit coupons are now printed before full parts

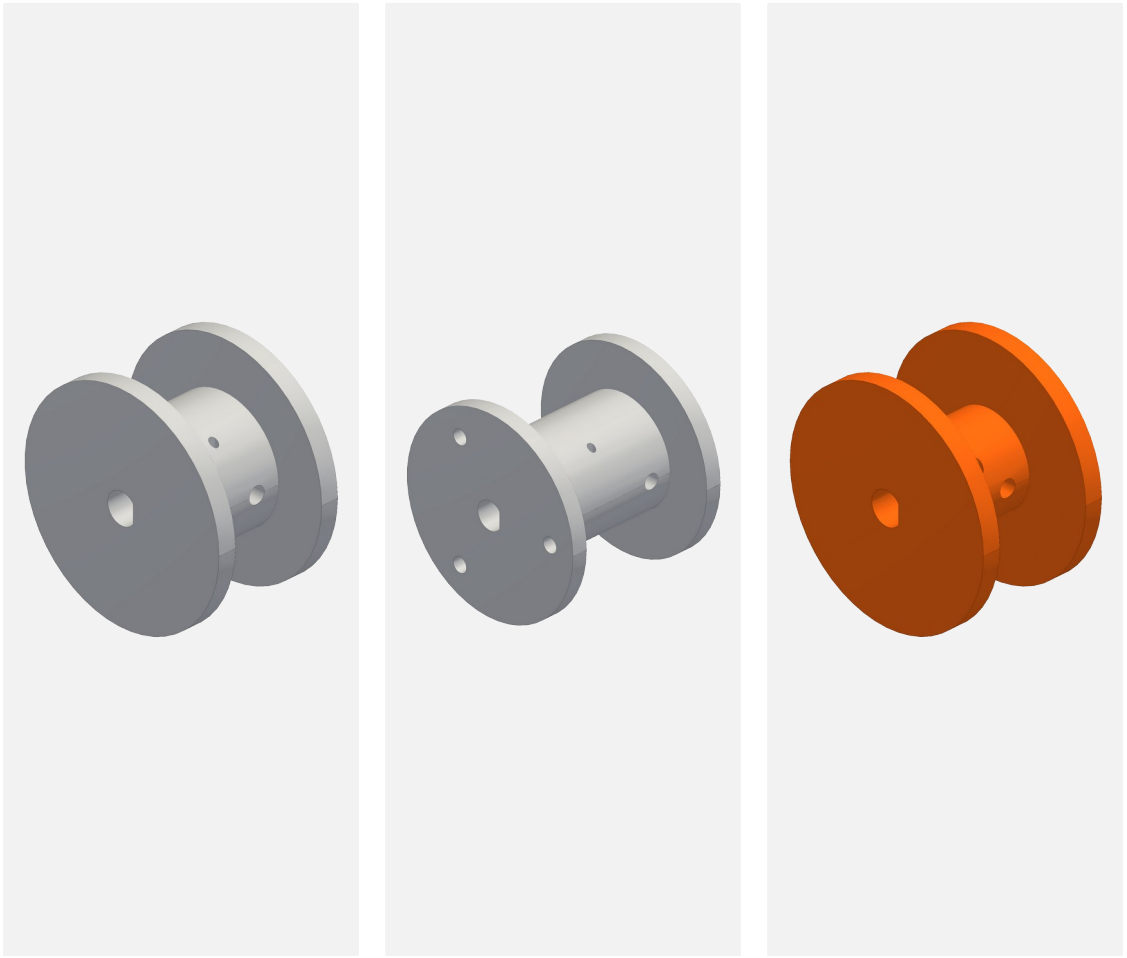
**Spool bore clearance**

$$c = (D_{hole} - u) - D_{shaft}$$

Ø5.2 bore too tight, so $u \geq 0.2$ mm

$c = (5.45 - 0.2) - 5.00 =$ **0.25 mm**

**Where:** c = clearance; u = printer undersize; $D_{shaft}$ = 5.00 mm motor shaft; version 3 bore Ø5.45 with a D-flat for the set screw



Spool versions 1, 2 and 3 (CAD)

# Heat Fin Manufacturing

- Shop drawing printed 1:1 and used as a hole template
- Prototype cut from 0.019 in aluminum sheet already on hand
- All five holes drilled in one mill setup to ±0.1 mm
- Fin bolted to the motor without reaming





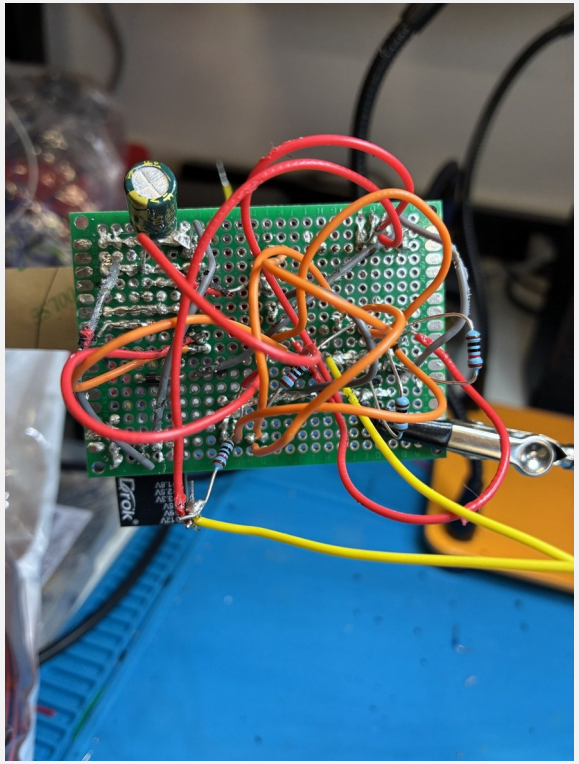





Shear and blanks, mill setup with the drawing, drilling, and the finished fin

# Electronics Bring-Up

- Driver stays disabled at boot, so a reset cannot move the motor
- Schematic generated from the soldered board, 16 nets
- First motor failed to turn: logic rail measured 1.14 V instead of 3.3 V
- Supply rails are now checked with a meter before any code is loaded





Soldered station board, left; bench bring-up, right

# First System Test

- Three stations: the fewest cables that fully constrain the claw
- Sequence: self-check, calibration, short move, then pickup
- 1 lb test load first, then one garment
- Pass limits defined before the run

*[Photo or video from the three-station test]*

| Check | Pass limit | Result |
|---|---|---|
| **Startup self-check** | All stations answer; slack taken up | *[fill in]* |
| **Calibration fit** | Held-out poses inside the safety margin | *[fill in]* |
| **Short move, 1 lb** | 300 mm at 50 mm/s, ≤ 2 mm per cable | *[fill in]* |
| **Pickup** | One garment into the hamper | *[fill in]* |

# Thermal Verification

- PETG mount must stay below 60 °C
- Hand calculation: 56–59 °C with the prototype fin, 45–49 °C with the 1/8 in fin
- SolidWorks thermal study of the motor, fin, screws and mount
- 90-minute hold test at 1.5 A to confirm

*[SolidWorks temperature plot: motor, fin, screws and mount]*

| Method | 0.019 in fin | 1/8 in fin | PETG mount |
|---|---|---|---|
| **Hand calculation** | 56–59 °C | 45–49 °C | About motor temperature |
| **SolidWorks** | *[fill in]* | *[fill in]* | *[fill in]* |
| **Hold test** | *[fill in]* | *[fill in]* | *[fill in]* |

Motor temperature at 1.5 A, 23 °C room, vented shell

# Good and Poor Decisions

| Good decision | Outcome |
|---|---|
| **One controller per winch** | Five times less wire; each station tested on its own |
| **Slide-in wall mount** | A station comes off while its anchor stays in place |
| **Camera calibration instead of bead homing** | Whole room reachable; no tape measure needed |
| **Gear strength checked at servo stall** | Module 1 gears would have failed; fixed in CAD before printing |

| Poor decision | Outcome |
|---|---|
| **Mass budget set late** | Claw and jeans are 0.2 kg over capacity, even at 1.5 A |
| **Bead homing as the first plan** | Under a third of the room reachable; redesign needed |
| **Spool printed before a fit test** | Three prints to fit the motor shaft |
| **Several stations built before one was proven** | Every fix repeated on each station *[confirm]* |

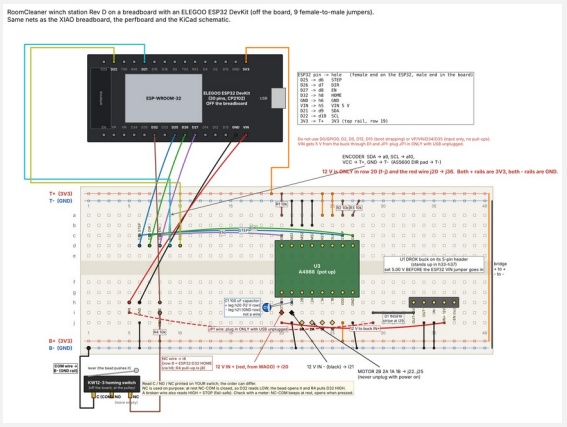
# Thorough vs. Scrappy

## Thorough



- Heat fin sized by calculation and machined in one mill setup to ±0.1 mm
- Station schematic checked net by net against the soldered board
- Gear strength checked at servo stall before printing

## Scrappy



- Failed controller replaced by an ESP32 dev board on nine jumpers, same firmware
- Floor board made from paper tags on a 250 mm grid
- Prototype fin cut from 0.019 in sheet already on hand

**Approach: thorough where a mistake is expensive or unsafe, scrappy where it is cheap to undo.**

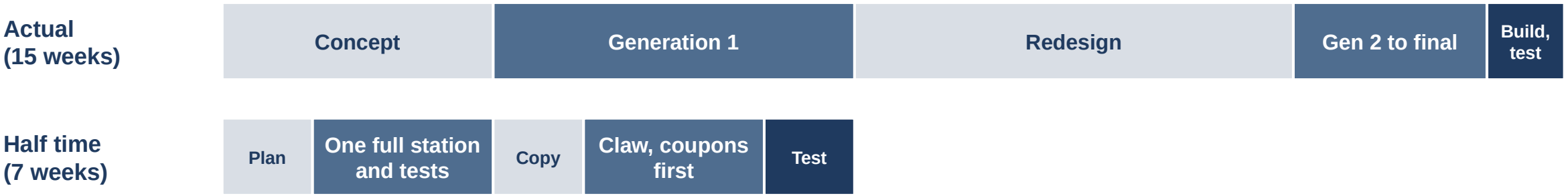
# Lessons Learned

## What I would do differently

- Set the mass budget early, from the measured winch pull
- Print fit coupons before full parts
- Prove one complete station before building the rest
- Check supply rails with a meter before debugging code

## Completing it in half the time

- Freeze requirements in the first week
- Order long-lead parts on day one
- Start with three stations
- Limit each part to one generation, gated by a test




Same time scale. Actual schedule: July 16 to October 27.

# Questions

Jasper Buntinx | jasper_buntinx@tamu.edu

# Appendix

Supporting detail for questions

# Project Status

| Part | Status | Next test |
|---|---|---|
| **Wall stations** | Built; bench-commissioned *[how many]* | Three-station test, then the fourth |
| **Station electronics** | Built; drivers at 1.5 A *[how: Vref reading]* | 90-minute hold test at 1.5 A |
| **Heat fin** | Prototype 0.019 in built; 1/8 in designed | Machine the 1/8 in fin; hold test |
| **Thermal model of the station** | Hand calculation at 1.5 A; SolidWorks model set up | Run it; then the hold test |
| **Claw body: tube, mount, servo, gears** | Built | Weigh it |
| **Claw hands, camera housing, encoder casing** | Checked in CAD; printing | Fit checks on the first print |
| **Calibration and control software** | Simulated end to end | First hardware run |
| **Winch pull** | Calculated at 1.5 A: 25–34 N | Luggage-scale pull test |

# Fin Sizing for 4 W

- Goal: the fin alone carries all 4.05 W
- 0.019 in sheet cannot at any length; maximum about 2 W
- 1/8 in plate needs 145–185 mm
- Thickness matters more than length

**Most heat any fin length can carry**

$$q_\infty = w\sqrt{(2\,h\,k\,t)} \times \theta_b$$

$$0.019\text{ in: } 0.044 \times \sqrt{(2 \times 9.3 \times 167 \times 0.000483)} \times 37$$

$$= \mathbf{2.0\ W} < 4.05\ \text{W}$$

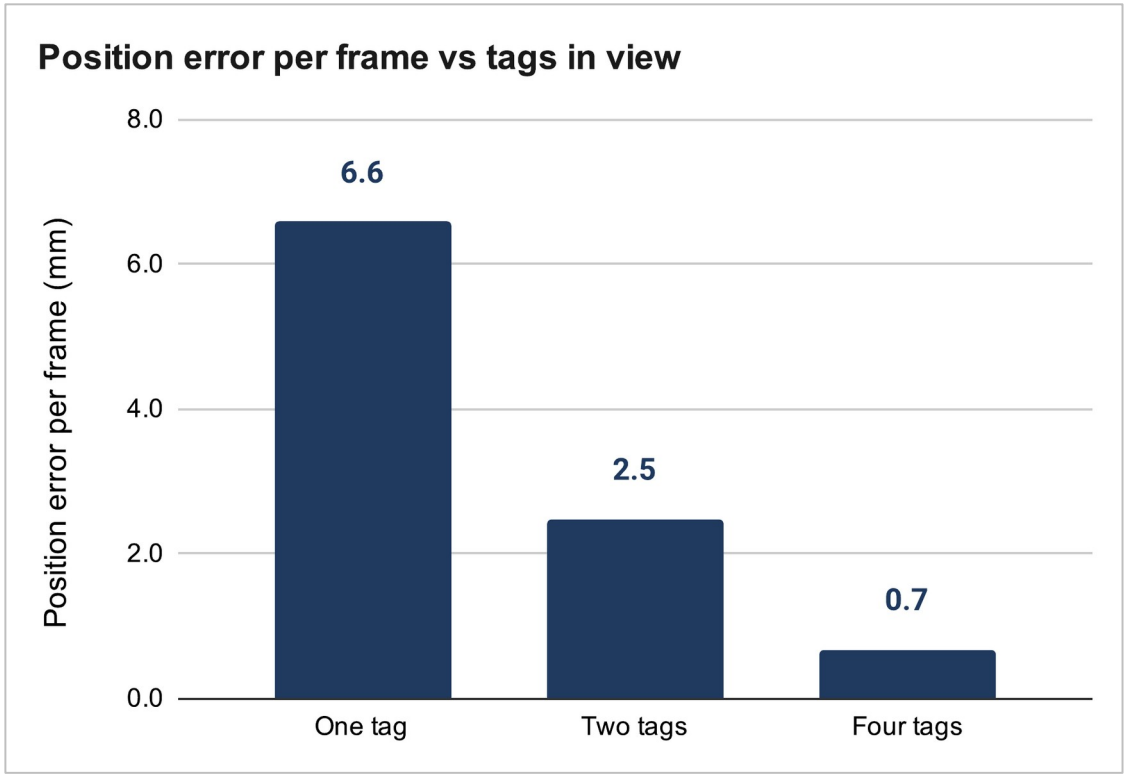
**Where:** w = 0.044 m fin width; h = 9.3 W/m²·K, painted; k = 167 W/m·K, 6061 aluminum; t = 0.000483 m; $\theta_b$ = 60 − 23 = 37 K

| Painted 6061, 44 mm wide | Mass | Motor at 200 mm, fin alone | Shortest for ≤ 60 °C |
|---|---|---|---|
| **0.019 in (prototype)** | 14 g | 100–120 °C | None (2.0 W max) |
| **1/16 in** | 47 g | 66–72 °C | None (3.6 W max) |
| **0.090 in** | 68 g | 60–65 °C | 210 mm or none |
| **1/8 in (design)** | 94 g | 56–59 °C | 145–185 mm |
| **3/16 in** | 141 g | 53–55 °C | 125–140 mm |
| **1/4 in** | 189 g | 51–52 °C | 115–125 mm |

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

# Floor Board Accuracy

- Position error equals angle error times camera height
- One tag in view: 6.6 mm; four tags: 0.7 mm
- Averaging removes noise, but not bias




Source: RoomCleaner engineering calculations, “Floor board” (calculated values)

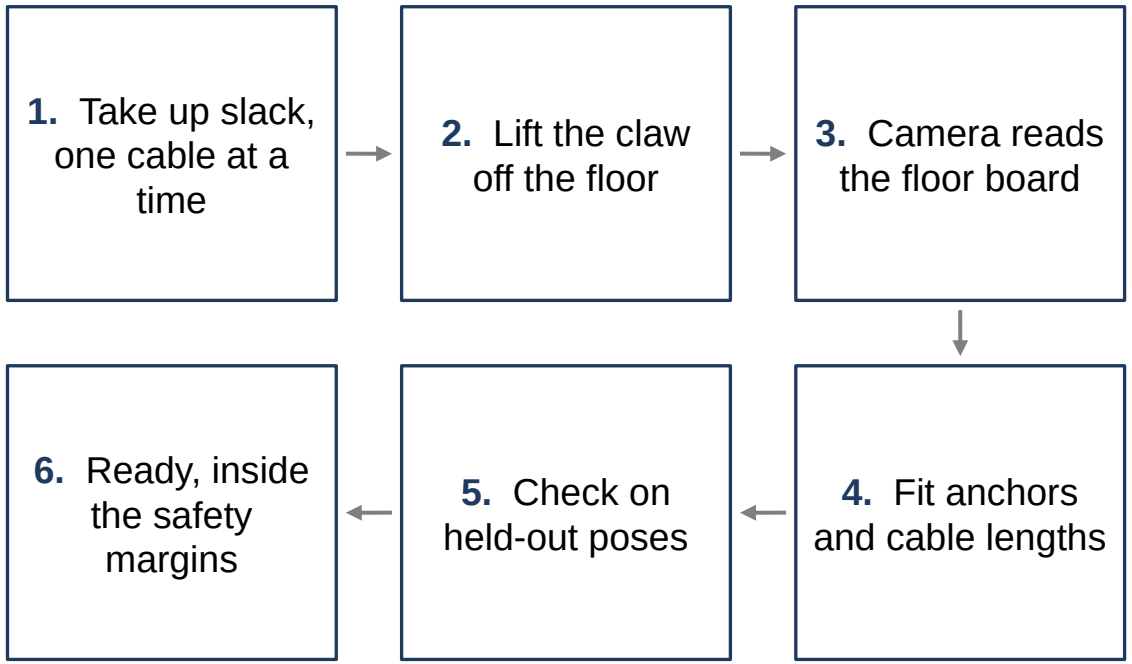
**Position error with one tag in view**

$$\delta x = \sigma_\theta \times L = 12.9\ \text{mrad} \times 0.51\ \text{m} = \mathbf{6.6\ mm}$$

**Where:** $\sigma_\theta$ = pose-angle noise with one tag in view; L = 0.51 m camera height above the board

# Startup Calibration

- No tape measure needed
- Fits anchor positions and cable zero lengths
- Held-out poses check the fit
- Full room allowed only within the fit's error bars




**Least-squares calibration fit**

$$\min \Sigma_k \Sigma_i \left( L_{i0} + d_{ik} - \|\mathbf{a}_i - \mathbf{p}_k\| \right)^2$$

$$\text{Full room only if } k\,\sigma \le \text{safety margin}$$

**Where:** $a_i$ = anchor i; $p_k$ = claw pose k from the camera; $d_{ik}$ = cable paid out, from the encoder; $L_{i0}$ = cable zero length; σ = fit error; k = confidence multiplier

# Control Logic: Generation 1 vs. Final

| | Generation 1 | Final |
|---|---|---|
| **Where is the claw?** | Assumed: where the cables were told to put it | Measured: claw camera on the floor board, refit at every startup |
| **Room geometry** | Point mass; anchors from a tape measure | Rigid claw; anchors and cable lengths fitted, with error bars |
| **Who runs the motors** | One central board steps all four | Each station runs time-stamped moves; the laptop coordinates |
| **Finding laundry** | Ceiling camera, pixel to floor map | Claw camera, ray from the claw's pose to the floor |
| **Did the grab work?** | Assumed once the claw closed | Camera checks pickup and delivery |
| **Faults** | Emergency stop only | Supervisor: a stop keeps torque; nothing restarts by itself |

# Operating Sequence

**1. Start up**

- Self-check
- Take up slack
- Lift a few centimeters
- Calibrate
- Set the trust level

→

**2. Find**

- Rise to scan height
- Detector lists the items
- Map each to the floor
- Choose the nearest

→

**3. Pick and deliver**

- Hover until still
- Descend and close
- Camera confirms the grip
- Move to the hamper and release
- Camera confirms the drop

→

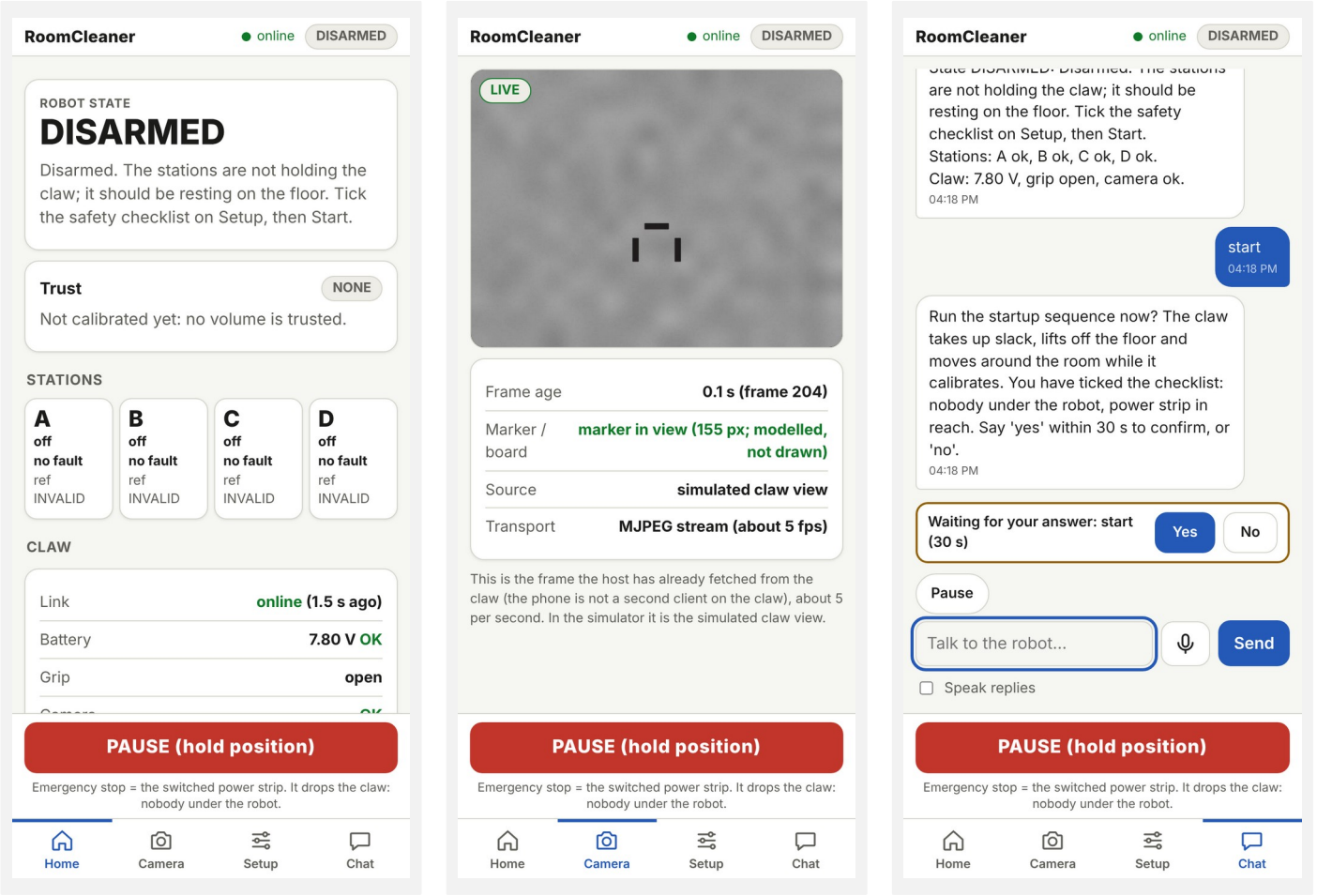
**4. Put away**

- Lower over the floor board
- Camera confirms the floor
- Disarm after a typed confirmation

- After each delivery the robot returns to step 2; three failures in a row pause it for the operator
- PAUSE or any fault stops every station at the same point on the path, still holding; nothing restarts by itself

# Phone App

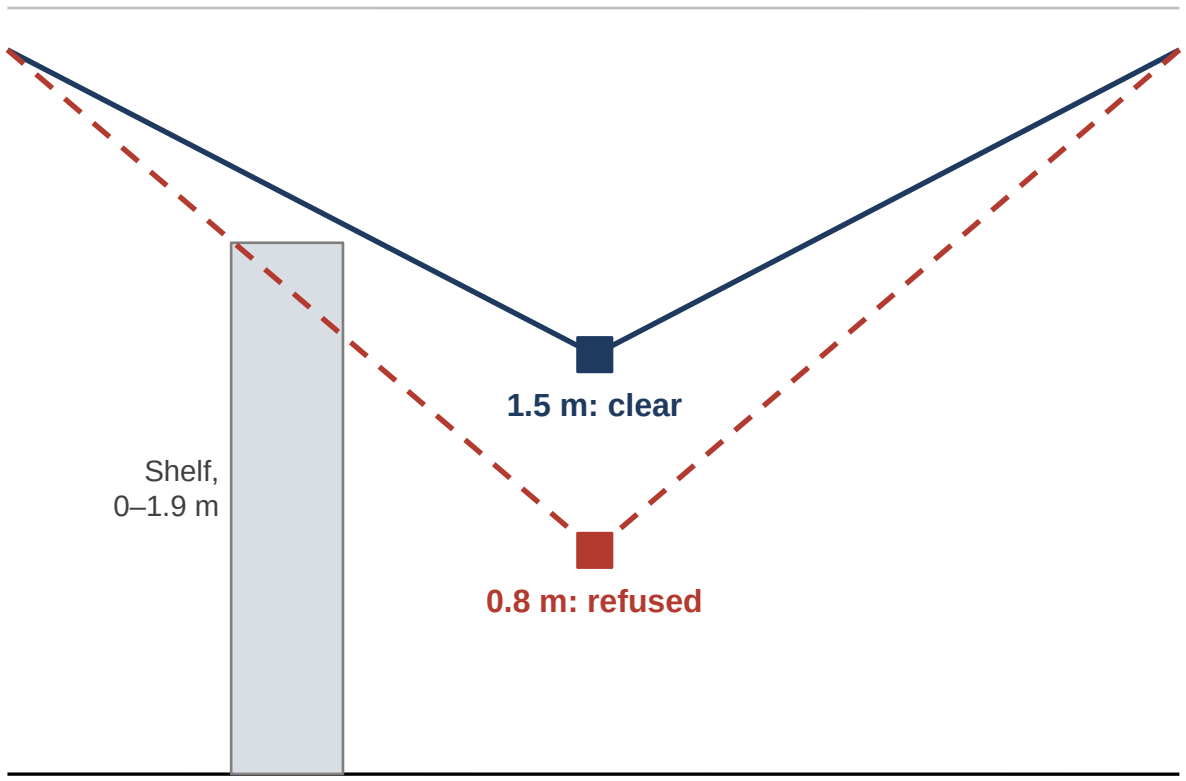
- Shows status, the live claw camera and guided setup
- Every motion asks for confirmation first
- PAUSE holds position immediately
- The switched power strip is the hard stop
- A reboot never resumes a task




Phone app: home, camera and chat screens

# Obstacles

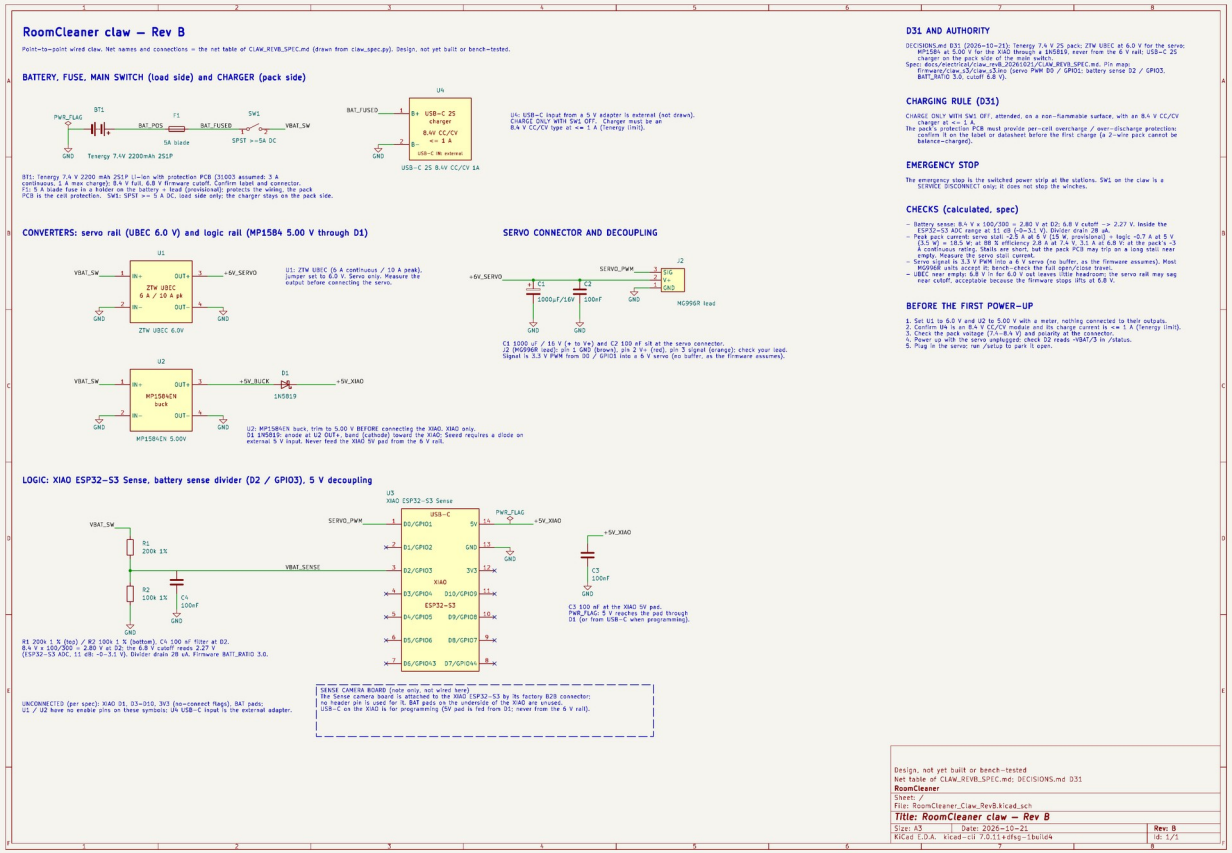
- Known obstacles modeled as keep-out cylinders
- Every cable checked at every point of the path
- Moves that fail the check are refused before any motor turns
- People are not detected yet, so nobody stands underneath




Side view along a room diagonal: the same floor point at two heights, with an example shelf

# Claw Electrical

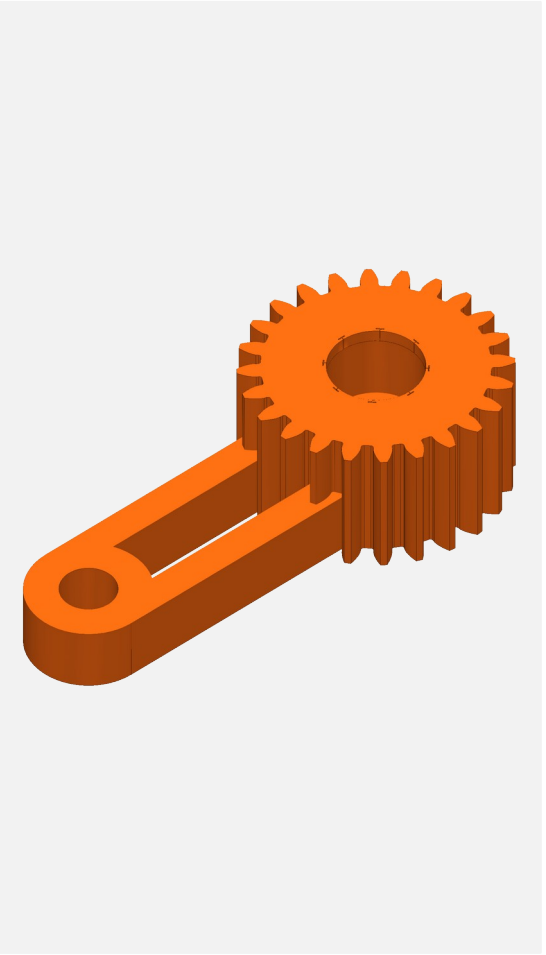
- Two-cell lithium-ion pack: 7.4 V with a 5 A fuse
- 6 V servo rail and a separate 5 V logic rail
- Peak draw of 2.8–3.1 A against a 3 A pack rating
- Charged with the switch off, attended, at 1 A or less



# Gear Strength

- Module 1.5 gears; PETG allowable stress 22.5 MPa
- Safety factors at 6 V stall: 2.05 pinion, 2.50 compound gear, 2.00 sync gear
- Module 1 gave safety factors of 0.35–0.41 and was rejected





**Lewis bending stress at servo stall**

$$F_t = 2\tau / d = 2 \times 1.078 / 0.027 = 79.9\ \text{N}$$

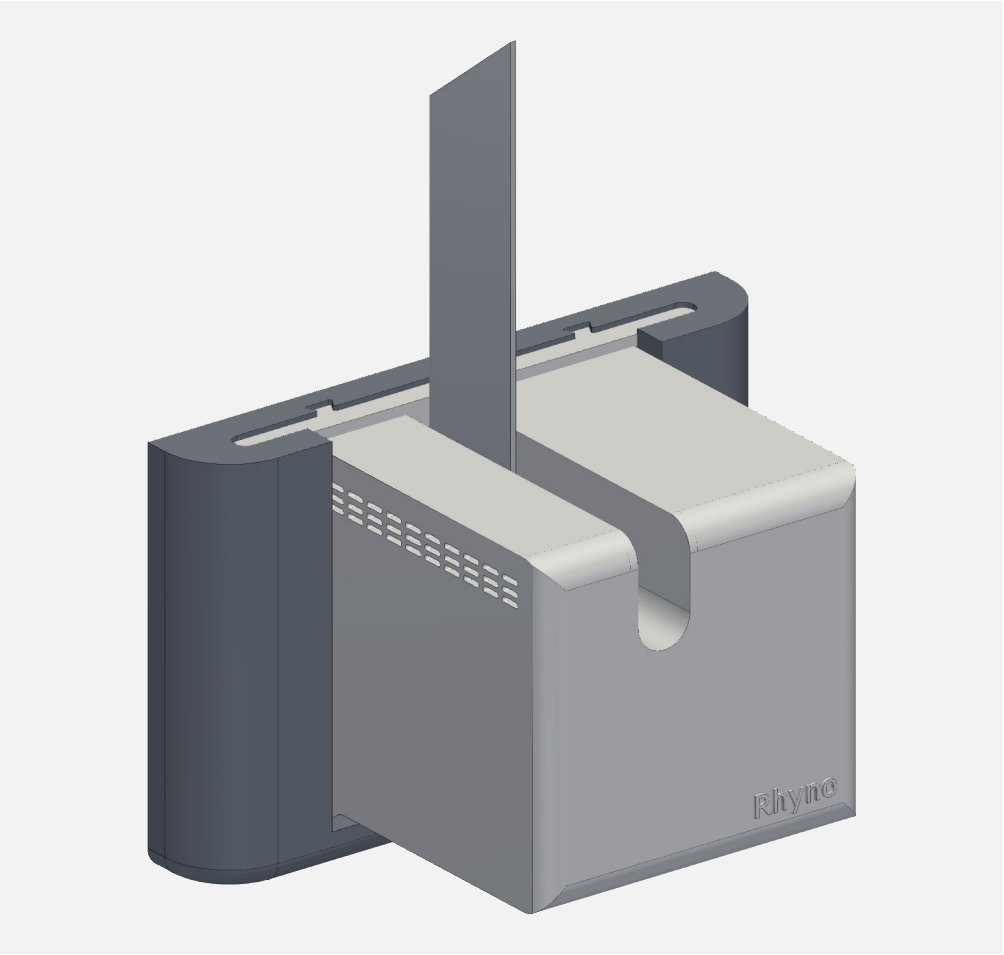
$$\sigma = F_t / (b\ m\ Y) = 79.9 / (15.7 \times 1.5 \times 0.309) = 11.0\ \text{MPa}$$

$$SF = \sigma_{allow} / \sigma = 22.5 / 11.0 = \mathbf{2.05}$$

**Where:** τ = servo stall torque; d = pitch diameter (18 teeth × 1.5 mm); b = face width; m = module; Y = Lewis form factor; SF = safety factor

# Safety

- One hard stop: the switched power strip
- Faults and PAUSE hold motor torque
- Anchors in wall studs; nobody stands under the robot
- Motors are never unplugged under power

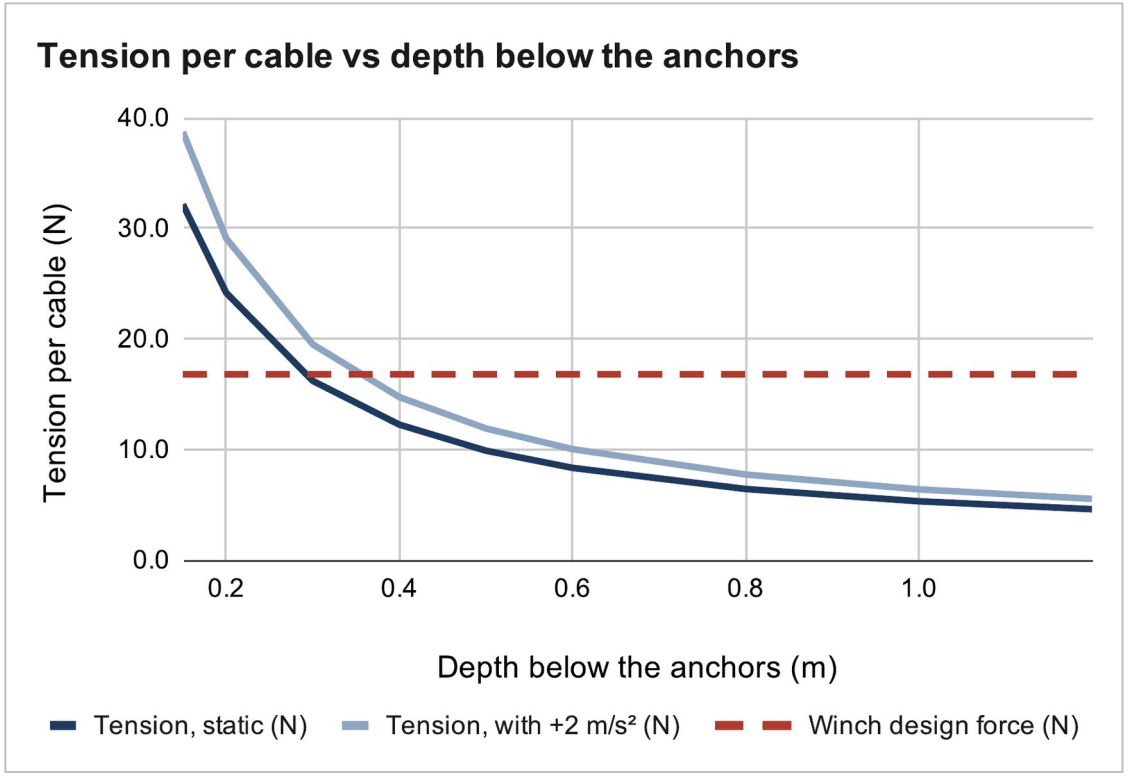


# Acronyms

| Term | Meaning |
|---|---|
| **CAD** | Computer-aided design |
| **PETG** | Polyethylene terephthalate glycol (the printed plastic) |
| **TPU** | Thermoplastic polyurethane (flexible printed plastic) |
| **NEMA 17** | National Electrical Manufacturers Association motor frame, 1.7 in square |
| **I²C** | Inter-Integrated Circuit: a short-range two-wire sensor bus |
| **ESP32** | A Wi-Fi microcontroller family (Espressif) |
| **A4988, AS5600** | Stepper-motor driver chip (Allegro); magnetic angle sensor (ams) |
| **YOLO** | “You Only Look Once”: an object-detection model family |
| **SF** | Safety factor: allowable stress divided by actual stress |

# Calculations and Sources

- All charts come from my engineering calculation sheet
- Each result is a formula on labeled inputs
- Sources:
  - Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
  - Allegro MicroSystems, A4988 datasheet




Source: RoomCleaner engineering calculations, “Cable tension” (calculated values)

RoomCleaner engineering calculations (Google Sheets)